Toward Independent Outdoor Running for People Who Are Blind or Have Low Vision: Final Report

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Abstract

People who are blind or have low vision face major barriers to independent outdoor running because most assistive tools are designed for walking and still rely on sighted guides. In this project, we explored how a wearable system with real-time sensing and haptic feedback could better support independent running. We first interviewed two blind runners to understand their current practices, safety concerns, and opinions about wearable guidance. Based on those insights, we built three prototypes, moving from camera-based obstacle detection to a wrist-worn haptic device. We then tested the latest prototype in a pilot study with two blindfolded sighted participants using stationary signal-recognition tasks, live indoor navigation, and follow-up interviews. Our results showed that left and right directional signals were easier to understand than non-directional signals such as slow down and obstacle warning. We also found that vibration remained noticeable during movement, although user trust in the system was still only moderate. Overall, our findings suggest that wearable haptic guidance is a promising direction, but the system still needs clearer signal design and further testing with blind and low-vision users in real-world outdoor settings.

CCS Concepts

• **Human-centered computing** → **Accessibility systems and tools**; **Empirical studies in accessibility**; *User studies*; *Haptic devices*.

Keywords

Accessibility, Blind and Low Vision, Wearable Devices, Haptic Feedback, Vibrotactile, Assistive Technology, Running, Outdoor Navigation, Computer Vision, Human-Centered Design

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1 Introduction

Individuals who are blind or have low vision (pBLV) face significant barriers to independent physical activity, and outdoor running remains especially difficult because it requires fast reactions to dynamic obstacles and often depends on a sighted guide. Existing accessibility tools such as white canes, guide dogs, and audio navigation can support walking, but they do not fully address the demands of running, where movement speed, environmental variability, and safety risks are substantially greater [8, 15, 16, 19, 22]. As a result, many pBLV runners still rely on tether-based guided running, which improves safety but limits spontaneity, flexibility, and independence.

Recent work in HCI and assistive technology has made progress in wearable sensing, computer-vision navigation, and non-visual feedback systems, yet most existing solutions are designed for walking-speed mobility or controlled indoor settings rather than outdoor running [1, 9, 21]. Running introduces a distinct combination of shorter reaction windows, stronger environmental uncertainty, and greater dependence on hearing, meaning that systems adapted from pedestrian navigation may not transfer directly to high-mobility contexts.

In this project, we investigate the following research question: *How can a body-mounted wearable system, using real-time environmental sensing and non-visual feedback, be designed to safely, effectively, and satisfyingly support independent outdoor running for pBLV individuals?* This question is important because outdoor running is not simply a faster version of walking. It requires rapid responses to changing obstacles, reliable guidance during movement, and a form of feedback that does not interfere with the runner's ability to hear the surrounding environment. In other words, the challenge is not only technical, but also experiential: the system must help users move safely, understand feedback quickly, and feel confident enough to rely on it during active motion.

To explore this question, we developed the *Virtual Tether*, a wearable navigation concept inspired by the familiar guide tether used by blind runners. Rather than treating the problem as only a sensing problem or only a feedback problem, we adopted a human-centered process grounded in formative input from blind runners, iterative prototyping, and pilot evaluation. The idea behind the Virtual Tether was to preserve the trust and simplicity of tether-based communication while translating it into a wearable system that could support more independent navigation.

We further broke this research question into three parts. **RQ1 (Usability)** asks whether users can distinguish directional haptic patterns, such as left, right, and obstacle cues, during active movement. This helps us understand whether the output language itself is clear enough to support navigation. **RQ2 (Perceptibility)** asks

at what movement speeds vibration patterns begin to feel weaker, less reliable, or harder to identify. This addresses a key concern from our formative study, where participants predicted that vibration might become difficult to notice while running. **RQ3 (Trust)** asks what level of confidence users report in vibration-based guidance. This is important because even if a system is technically functional, it will not be useful in practice unless users feel that it is dependable enough to support real movement and decision making.

Together, these sub-questions let us evaluate the system from three complementary perspectives: whether the cues are understandable, whether they remain noticeable during motion, and whether users feel willing to trust them. Framing the project in this way allowed us to move from a broad design question about independent outdoor running to a more concrete investigation of the sensing, feedback, and user-experience challenges that shape assistive wearable design.

2 Related Work

2.1 Accessible Physical Activity and Guide Dependence

People who are blind or have low vision are often excluded from mainstream physical activity not because of lack of motivation, but because of structural barriers in design, environment, and access [2, 16, 22]. Prior work shows that many exercise systems assume visual interaction, controlled environments, or sighted assistance, which can make independent participation difficult for pBLV users. Running makes these constraints especially visible because it is not only a form of exercise, but also a form of autonomy and freedom of movement.

Many pBLV individuals want to run for health, stress relief, and a sense of normalcy, yet outdoor running almost always depends on a sighted guide or tether [1, 19]. This dependence creates scheduling burdens, limits spontaneity, and reduces the experience of free, independent movement. Even systems that improve coordination between guides and runners, such as haptic sharing approaches, still rely on another person and therefore do not fully solve the independence problem [7]. Together, this body of work shows that the key challenge is not only supporting mobility, but doing so in a way that reduces dependence on human guidance while maintaining safety and confidence.

2.2 Environmental Sensing and Wearable Navigation

A large body of assistive-technology research has focused on environmental sensing for navigation, including smart canes, wearable cameras, and computer-vision systems that detect obstacles, pedestrians, and path boundaries [10, 11, 13]. These systems demonstrate that real-time scene understanding for navigation is becoming increasingly feasible, especially as computer-vision models improve in speed and accuracy. They are valuable because they extend environmental awareness beyond the limited range of the white cane and can provide users with earlier warnings about hazards.

However, most such systems are still designed around walking-speed use, a single forward-facing view, and relatively predictable environments. Prior work has also noted that many assistive technologies are evaluated in controlled settings, which limits understanding of how they perform under real-world variability [4, 5]. These assumptions become more problematic for running, where lateral awareness, rapid response, and environmental unpredictability are much more important. A runner may need to react within a much shorter time window than a pedestrian, and a single missed detection or delayed command can have greater safety consequences. This suggests that sensing systems for running must be more responsive and more robust than systems intended only for walking.

2.3 Non-Visual Guidance for Active Running

Non-visual guidance systems mainly use either audio or haptic feedback. Audio guidance, such as spoken instructions or spatial sound, can help users navigate complex spaces, but it can also block important environmental sounds like traffic, footsteps, or nearby voices [6, 18]. For running, this limitation is especially important because pBLV runners often depend heavily on hearing to maintain situational awareness. Haptic guidance avoids this problem and has been effective for walking-speed navigation, especially when signals are mapped clearly to the body [21, 23]. Force-feedback and skin-stretch systems go a step further by creating a pulling sensation, which may feel more natural for running because it is closer to the experience of following a tether [9, 12].

Some studies have started to explore running more directly. Drone-guided and robotic systems show that non-human guidance is possible, and some haptic running aids have improved performance in controlled settings such as indoor tracks [1, 9, 24]. At the same time, prior work still leaves several important gaps. Many running-oriented systems have only been tested in structured environments, with small participant samples, or with blindfolded sighted users rather than pBLV runners [20, 21]. In addition, many feedback designs are adapted from walking contexts and may not fully address the perceptual and cognitive demands of active running. Overall, there is still a need for systems that combine user-centered design, wearable sensing, and non-visual feedback specifically for independent outdoor running. Our work addresses this gap through formative interviews, iterative prototyping, and evaluation of an integrated wearable navigation system.

3 Research Method

To address our research question, we adopted a human-centered, iterative design approach combining qualitative formative research with iterative prototyping and pilot evaluation. Our methodology unfolded across three interconnected phases.

Phase 1: Formative Study. We conducted semi-structured interviews with two totally blind runners (P1 and P2) recruited through the Dark Running Group, a community organization that organizes guided running activities for people with visual impairments. Each interview lasted approximately 60–90 minutes and explored current running practices, the role of the tether (*peipaosheng*) as a communication channel between guide and runner, environmental hazards, trust development, and openness to assistive technology. Interview data were transcribed verbatim and analyzed using inductive thematic analysis [3], with team members independently coding transcripts before converging on shared themes.

From these findings, we synthesized six design requirements (DR1–DR6) that guided subsequent prototyping. First, the system should preserve familiar navigation cues by replicating the compact directional vocabulary already used through the physical tether. Second, it should support non-visual interaction so that feedback remains perceivable without requiring visual attention. Third, it should avoid heavy dependence on audio output, since spoken guidance may mask important environmental sounds during running. Fourth, it should minimize cognitive load by keeping feedback simple, lightweight, and easy to interpret during movement. Fifth, it should provide real-time responsiveness so that navigation cues arrive quickly enough to support continuous movement and obstacle avoidance. Finally, it should support trust and safety by prioritizing predictability and confidence during navigation.

Phase 2: Iterative Prototyping. Guided by the design requirements, we developed three prototypes in sequence, each targeting a distinct sub-problem of the overall system. V1 and V2 explored the *sensing* substrate—using a single forward-facing camera with computer vision to translate the visual scene into directional commands. V3 explored the *output* substrate—using a wrist-worn vibrotactile actuator to deliver these commands without requiring vision. Each prototype was evaluated informally to surface limitations, and these limitations directly motivated the next iteration. The three components were finally combined into an integrated end-to-end system that connects vision input to haptic output.

Phase 3: Pilot Evaluation. To validate core interaction principles before deployment with pBLV users, we conducted a controlled pilot study with sighted participants under blindfolded conditions across three movement states (seated, walking, light jogging). The pilot used a mixed-methods design combining quantitative pattern-recognition accuracy and response-time data with qualitative semi-structured interviews. Detailed methodology and findings are reported in Section 5.

This three-phase structure—*understand, build, test*—reflects established practice in accessibility-focused HCI research [14]and allowed us to ground each design decision in empirical evidence from the target population while iteratively de-risking the technical components of the system.

4 System and Prototype Design

We developed the *Virtual Tether* system through three iterative prototypes that progressively refined both the sensing substrate (computer vision) and the output substrate (haptic feedback), as shown in Figure 1. Rather than treating the system as a single monolithic artifact, we deliberately decomposed the design problem so that each prototype addressed a distinct technical question motivated by the formative study. This section describes the design rationale and implementation of each prototype, followed by the integrated end-to-end system.

4.1 Design Goals

Our formative study revealed that pBLV runners rely on a remarkably compact and trusted communication vocabulary mediated by the tether: subtle directional pulls for left and right corrections, and a sharp tug for emergency stops. Rather than introducing a new feedback language, we framed the design problem as *preserving this existing trusted interaction under a new sensing-and-feedback substrate*—a concept we term the *Virtual Tether*. Concretely, this framing translated to four design goals.

G1: Preserve the three-signal vocabulary. The system must produce *left*, *right*, and *stop* commands that map cleanly onto the tether's established signals (DR1).

G2: Provide an accessible output channel. Output must be perceivable without vision and ideally without sound, given participants' concerns about audio masking in outdoor environments (DR2, DR3).

G3: Operate in real time. End-to-end latency from sensing to feedback must support running speeds, with responsive cues delivered before the runner reaches the obstacle (DR3).

G4: Minimize cognitive load. Output should be minimal and well-timed rather than continuous narration, in line with formative findings that single-target cues are more usable than information-dense streams (DR4).

4.2 Prototype V1: Establishing the Computer-Vision Baseline

V1 was designed to test a single question: can real-time computer vision recover the tether's three-signal vocabulary from a single visual stream?

4.2.1 Implementation. V1 used a single forward-facing camera as input and applied a YOLO-based object detector [17] to each frame. For every detection, the system recorded three pieces of metadata: the object class, the horizontal position of its bounding box (binned into *left*, *center*, or *right*), and a coarse distance proxy derived from bounding-box size. The decision module prioritized the *closest* obstacle (largest bounding box) and emitted one of three commands—*left*, *right*, or *stop*—explicitly mirroring the tether vocabulary. The single-target focus operationalized the formative finding (G4) that minimal, well-timed cues are more effective than continuous narration. An experimental semantic-segmentation module was also explored for walkable-path detection.

4.2.2 Identified Limitations. V1 validated basic feasibility but exposed several limitations that directly motivated V2:

- The walkable-path segmentation module produced inconsistent estimates and could not reliably support recovery from off-path drift.
- Bounding-box size alone proved insufficient as a distance proxy: small near objects and large distant objects produced indistinguishable signatures.
- The centered-obstacle case lacked a principled tie-breaking rule, producing arbitrary directional decisions.
- Detection was sensitive to misclassification and missed small obstacles.
- Most importantly, V1 produced *no accessible output*—the system relied on visual debug overlays, precluding evaluation with non-sighted users.

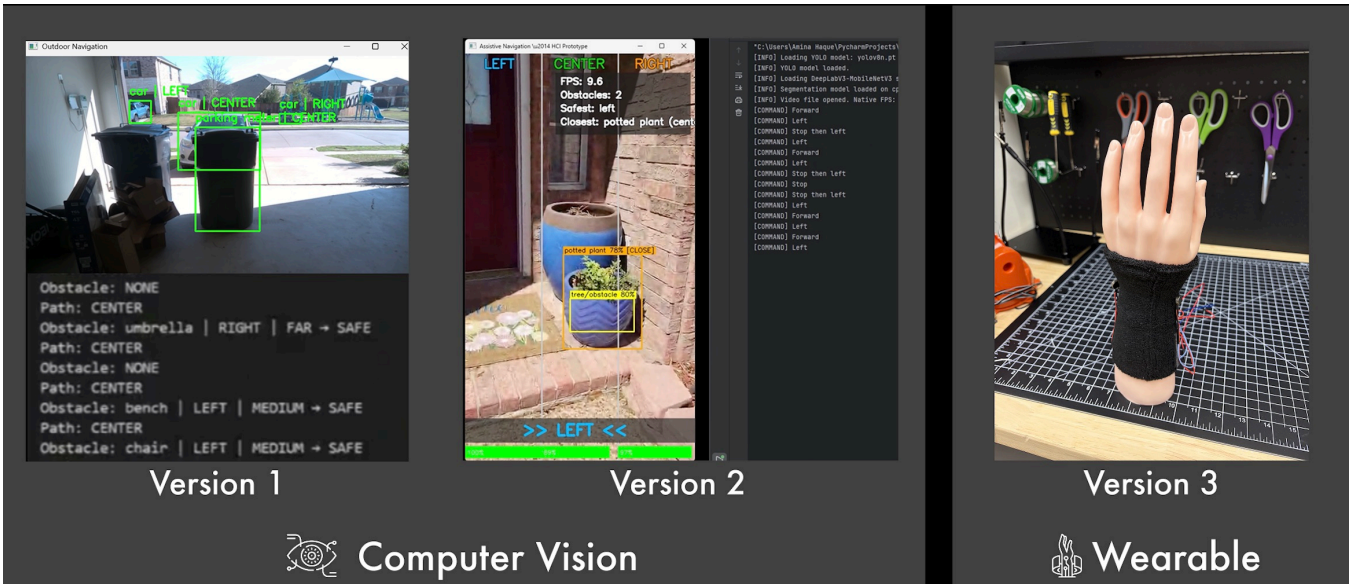


Figure 1: Three prototype iterations of the *Virtual Tether* system. Version 1 (left) establishes a computer-vision baseline using YOLO object detection with bounding boxes, region binning, and a coarse distance proxy; output is visual-only, limiting accessibility. Version 2 (center) introduces a modular five-stage pipeline—object detection, distance estimation, path analysis (bottom-40% open-space scoring across left/center/right zones), decision module, and text-to-speech output—adding accessible feedback and principled tie-breaking. Version 3 (right) shifts focus to the output substrate: a wrist-worn vibrotactile band built around an Arduino Uno and two coin vibration motors, designed to deliver the three-signal tether vocabulary silently and eyes-free. The integrated end-to-end system connects the V2 vision pipeline to the V3 wristband, replacing speech output with haptic cues.

4.3 Prototype V2: Modular Pipeline with Accessible Output

V2 restructured the system into a modular pipeline, refined the decision logic, and introduced an accessible output channel via text-to-speech.

4.3.1 Five-Module Architecture. We decomposed the system into five components, each independently iterable:

(1) **Object Detection.** A YOLO model restricted to runtime-relevant classes (people, vehicles, benches, bicycles, animals, and other scene-relevant obstacles).
(2) **Distance Estimation.** A three-level categorization (*far*, *close*, *very close*) computed from a combination of bounding-box size and vertical placement within the frame, providing a more reliable proximity signal than V1's size-only proxy.
(3) **Path Analysis.** An open-space scoring module that operates only on the bottom 40% of the frame—the region most likely to correspond to the immediate walking ground—and divides this region proportionally into left, center, and right zones, making the analysis robust across camera resolutions.
(4) **Decision Module.** A combined scoring rule that integrates obstacle proximity with open-space availability. The system prefers the forward direction unless one side offers meaningfully greater clearance, replacing V1's arbitrary tie-breaking with a principled comparison.
(5) **Voice Output.** A text-to-speech module that announces commands when they change and periodically re-issues them, removing the visual-monitoring requirement of V1.

4.3.2 Substantive Improvements over V1. Beyond modularity, V2 introduced several specific refinements: (1) a lighter segmentation backbone for real-time inference; (2) restriction of the segmentation *background* class as the only walkable region, reducing unsafe misclassification; (3) the bottom-40%-of-frame restriction described above; (4) proportional rather than fixed-pixel region thresholds; and (5) the open-space comparison logic that replaces V1's arbitrary tie-breaking. A visual debug overlay (color-coded bounding boxes, an openness bar graph, and a command banner) was retained for development purposes only.

4.3.3 Acknowledged Limitations. V2 still lacks true depth perception, has no temporal memory of moving objects, and offers only coarse directional granularity. It also relies on speech output, which formative participants flagged as problematic in noisy outdoor environments and during high-exertion running. These limitations were sufficient as a sensing substrate to test *alternative output modalities*, which directly motivated V3.

4.4 Prototype V3: Vibrotactile Wristband

V3 shifted the focus from sensing to output, asking whether a wrist-worn haptic device could deliver the three-signal vocabulary in a

form that pBLV runners could perceive and act on without vision or audio.

4.4.1 Design Rationale: Vibration as Functional Validation. While our formative findings (DR2) suggested that force-based feedback would ultimately be necessary to fully replicate the tether's pulling sensation, we deliberately chose vibration motors for this iteration as a pragmatic first step. Vibration actuators are low-cost, readily available, require minimal supporting electronics, and enable rapid iteration on pattern timing and spatial encoding without the mechanical complexity of force-based actuation. V3's intent was therefore not to deliver a deployment-ready solution, but to validate three foundational questions: *Can wrist-mounted spatial cues distinguish left from right? Can multi-pattern vocabularies be learned with minimal training? What perceptual constraints does active movement impose on vibrotactile feedback?*

4.4.2 Hardware. The wristband consisted of: (i) two DC 3V coin-style vibration motors (10 mm diameter × 3 mm thickness, 12,000 RPM) positioned on the left and right sides of the wrist for spatial directional cues; (ii) an Arduino Uno microcontroller for pattern generation and motor control via PWM; (iii) a 9 V battery for power; (iv) current-limiting resistors (100–220 Ω) on each motor line to protect the Arduino pins; and (v) an elastic athletic wristband with hook-and-loop straps that kept the motors stable against the skin during arm motion. The complete assembly weighed under 40 g.

4.4.3 Vibration Pattern Design. We designed five distinct vibration patterns that mapped onto the navigation vocabulary (Table 1). Left and right turns shared the same temporal structure (three medium pulses) but activated different motors, leveraging spatial encoding to minimize cognitive load. The obstacle warning combined rapid pulsing and bilateral activation to convey urgency. The on-track and slow-down patterns used distinctly different temporal envelopes to remain unambiguous. Pattern timing was calibrated for perceptual clarity: 200 ms pulse duration provides sufficient time for perception while remaining brief enough to avoid masking subsequent signals, and 100 ms inter-pulse gaps allow distinct beat perception.

4.4.4 Software Implementation. We developed Arduino firmware that exposes a simple keyboard-triggered Wizard-of-Oz interface over the serial port: each key press (`L`, `R`, `O`, `T`, `S`) triggers the corresponding pattern. This interface enabled rapid pattern triggering during user testing, allowed an experimenter to deliver cues in real time without requiring the full computer-vision pipeline, and supported decoupled evaluation of the output substrate independently of the sensing substrate.

4.5 Integrated System: Connecting Vision to Haptic Output

The final stage of our prototyping work integrated V2 and V3 into a unified end-to-end pipeline. V2's decision module—which had previously emitted commands as text-to-speech utterances—was redirected to send commands over a serial connection to the V3 Arduino firmware. The Arduino received each command and triggered the corresponding vibration pattern on the wristband. The data flow is illustrated as:

```
Camera → V2 CV Pipeline → Decision → V3 Wristband
→ Runner
```

This integration realizes the *Virtual Tether* concept end-to-end: visual scene information is sensed and interpreted on one side of the system, and the resulting navigational intent is delivered to the runner through silent, eyes-free haptic cues that preserve the three-signal tether vocabulary established in formative work. The integrated system serves as the artifact evaluated in our pilot study (Section 5).

5 Evaluation

5.1 Study Execution

We recruited two sighted participants (P1, P2) and used blindfolds during all tasks to simulate non-visual reliance on haptic cues. Although our target users are BLV runners, blindfolded sighted participants are a standard surrogate for early-stage assistive-technology pilots; running an unvalidated prototype with BLV participants would be ethically inappropriate before basic feasibility is established. We treat findings as formative, not confirmatory.

We followed a specific study plan analysis: (1) a stationary blindfolded *pretest* of 20 randomized cues per participant (4 trials × 5 signal types: Left, Right, On Track, Obstacle, Slow Down), recording accuracy and response time; (2) a *live navigation session* where participants walked an indoor route while a head-mounted camera streamed to our obstacle-detection module, which issued real-time haptic commands; and (3) a semi-structured *post-study interview* including a 1–10 confidence rating.

5.2 Data Collection

Quantitative. Pretest per-trial accuracy and response time were logged in a spreadsheet by the experimenter. Live-session metrics (duration, command counts, obstacle-frame counts, FPS) were logged automatically by our Python backend. Confidence ratings were captured during the interview.

Qualitative. We collected think-aloud comments, post-study semi-structured interview responses (signal clarity, perceptibility under movement, trust, design suggestions), and experimenter observations. Sessions were audio-recorded with consent and transcribed for coding.

5.2.1 Quantitative Analysis. Below are our results recorded in the tables.

Table 1: Vibration pattern design for the V3 wristband. Each pattern maps onto a navigational signal in the tether vocabulary; left/right turns share temporal structure but differ in spatial activation, while urgency-related signals (obstacle warning, slow down) are differentiated through pulse rate and duration.

Pattern	Motors	Intensity (PWM)	Temporal Structure
Left Turn	Left only	Medium (150/255)	3 pulses, 200 ms on / 100 ms gap
Right Turn	Right only	Medium (150/255)	3 pulses, 200 ms on / 100 ms gap
Obstacle Warning	Both	Strong (220/255)	6 pulses, 100 ms on / 50 ms gap
On Track	Both	Gentle (80/255)	Continuous, 1,000 ms
Slow Down	Both	Medium (150/255)	4 pulses, 400 ms on / 200 ms gap

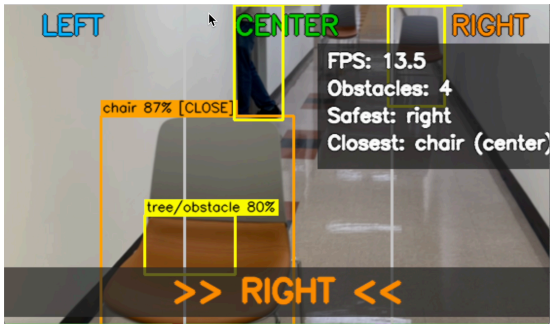


Figure 2: Live navigation session: blindfolded participant wearing the V3 wristband and the head-mounted obstacle-detection camera.

Table 2: Pretest Response Time (s)

Metric	P1	P2
Mean / Median	4.27 / 4.58	5.58 / 5.01
SD	1.35	3.89
Min / Max	2.25 / 6.42	2.26 / 18.10
Skewness	−0.02	+1.92

Table 3: Live Session Metrics

Metric	P1	P2
Session duration	129.6 s	97.9 s
Total commands	37	39
Fwd / L / R / Stop	17/9/6/5	14/11/4/10
Frames w/ obstacle	70.3%	92.3%
FPS Mean / SD	10.64 / 1.52	14.44 / 9.10

Table 4: Pretest Accuracy by Signal Type

Signal	P1	P2	Pooled
Left Turn	4/4 (100%)	4/4 (100%)	100%
Right Turn	2/4 (50%)	4/4 (100%)	75%
On Track	3/4 (75%)	3/4 (75%)	75%
Obstacle Warning	4/4 (100%)	1/4 (25%)	62.5%
Slow Down	1/4 (25%)	3/4 (75%)	50%
Overall	**70.0%**	**75.0%**	**72.5%**

The descriptive statistics show that both participants were generally able to perceive and respond to the haptic signals, with pooled pretest accuracy reaching 72.5%. Directional cues achieved the strongest performance, with Left Turn reaching 100% accuracy and both Right Turn and On Track reaching 75%, while Slow Down showed the weakest performance at 50% and Obstacle Warning reached 62.5%. This shows that there may have been a greater confusion with the non-directional temporal patterns.

During the live sessions, participants successfully completed navigation tasks lasting 129.6 s and 97.9 s while receiving 37 and 39 commands respectively, despite obstacles appearing in 70.3% and 92.3% of frames. This shows that the system remained functional and perceptible during active movement.

5.3 Inferential Statistics

RQ1 – One-sample t-test per signal vs. chance (0.20). RQ1 asks whether each pattern is recognizable, not whether two groups differ. The meaningful comparison is each signal's accuracy against the fixed chance baseline (1/5 = 0.20). A paired/independent t-test would require two groups; ANOVA is unnecessary because each signal is tested against the same fixed benchmark. We test per-signal because RQ1 concerns *individual* pattern distinguishability.

Table 5: Accuracy vs. Chance (df=7)

Signal	Mean	t	p	Above chance?
Left Turn	1.000	∞	$< .0001$	Yes
Right Turn	0.750	3.361	.0121	Yes
On Track	0.750	3.361	.0121	Yes
Obstacle Warning	0.625	2.323	.0532	Marginal
Slow Down	0.500	1.587	.1564	No

Of the three signals RQ1 names, Left and Right Turn performed above the chance baseline in this pilot study, while Obstacle Warning marginally missed significance ($p = 0.053$). The prototype *partially* answers RQ1: directional cues work; obstacle warning needs improvement.

RQ2 Part A – One-sample t-test, overall accuracy vs. benchmarks. We test against two fixed values: chance (0.20) and a 75% HCI usability threshold. Both are reference values, not second groups, so a one-sample t-test fits. With overall accuracy M = 0.725 (SD = 0.452, n = 40): vs. chance, $t = 7.343$, $p < .0001$ (significant); vs. 75% threshold, $t = -0.350$, $p = 0.729$ (not significant). Vibration is perceivable but not yet operationally reliable.

RQ2 Part B – Paired t-test, stationary vs. moving response time. The same participants were measured under two conditions, so a within-subjects paired t-test is appropriate; an independent t-test would treat individual differences as noise. Stationary pretest M = 4.93 s (SD 2.95) vs. live moving session M = 2.33 s (SD 3.97); mean difference = 2.60 s, $t(39) = 3.574$, $p = 0.001$. Response time was *faster* during movement, contradicting the formative prediction that "vibration becomes imperceptible when running." Live navigation likely demands continuous attention to haptic cues, while the pretest allowed slow deliberation.

Code	P	Quote	Theme
1	P2	"The clearest is probably left and right."	Directional clarity
2	P1	"The left and right was very clear to tell."	Directional clarity
3	P2	"Sometimes it's a little confusing to know the signals that aren't right and left, like stop or obstacle in the way or slow down."	Non-directional confusion
4	P1	"The most confusing was the slow down and object close."	Non-directional confusion
5	P2	"No, I wouldn't say that there's any big difference other than maybe the vibrations happening more rapidly when you're actually going through it."	Perceptibility during movement
6	P1	"Without the sight, it made my touch more sensitive."	Heightened sensitivity without vision
7	P1	"I would feel confident if I wasn't moving too fast, but I think around maybe a 7."	Moderate Trust and Design Feedback
8	P2	"If I had a more complex path, maybe like a 6 or a 7."	Moderate Trust and Design Feedback
9	P2	"Maybe you could add an extra sensor for obstacle, like on the bottom of your hand, just so it doesn't conflict with the other sensors."	Moderate Trust and Design Feedback
10	P1	"I think most likely it was just the slow down and then the close object. So that was the main part."	Moderate Trust and Design Feedback

5.3.1 Qualitative Analysis. We performed open coding on the transcripts and grouped codes into four themes through iterative discussion among the three authors.

Theme 1: Directional Clarity. Both participants spontaneously named left and right turn as the clearest signals (P1: "the left and right was very clear to tell"; P2: "the clearest is probably left and right"). Spatially encoded vibration mapped intuitively onto direction without learning, indicating that motor placement on opposite sides of the wrist is immediately understood.

Theme 2: Non-Directional Confusion. Both participants independently named Slow Down and Obstacle Warning as the most confusing (P2: "it's a little confusing to know the signals that aren't right and left"; P1: "the most confusing was the slow down and object close"). The convergence indicates a design-level issue, not unfamiliarity, and aligns directly with the chance-level accuracy on these two signals.

Theme 3: Perceptibility During Movement. Neither participant reported vibration becoming imperceptible during walking. P2: "no…other than maybe the vibrations happening more rapidly when you're actually going through it." P1: "without the sight, it made my touch more sensitive." This contradicts the formative prediction and supports the paired-t-test finding that response time *improved* during motion.

Theme 4: Moderate Trust and Design Feedback. Confidence ratings clustered at 6–7/10, conditioned on speed and route complexity (P1: "I would feel confident if I wasn't moving too fast, but I think around maybe a 7"; P2: "if I had a more complex path, maybe like a 6 or a 7"). Participants offered concrete fixes, most notably P2's suggestion to relocate the obstacle motor: "add an extra sensor for obstacle, like on the bottom of your hand, just so it doesn't conflict with the other sensors," directly addressing the Theme 2 confusion.

5.3.2 Result. Bringing the quantitative and qualitative findings together yields four conclusions tied directly to our research questions. Across all four, the two data streams reinforce rather than contradict each other, which strengthens our confidence despite the small sample.

Spatial encoding works; temporal encoding does not (RQ1). Quantitatively, Left and Right Turn reached 100% and 75% pooled accuracy and were significantly above chance ($p < .0001$ and $p = .012$). Slow Down (50%) sat at chance and Obstacle Warning (62.5%) only marginally exceeded it ($p = .053$). The qualitative data converged on exactly the same split: both participants spontaneously named left/right as the clearest signals and Slow Down/Obstacle as the most confusing, using nearly identical language without prompting. The mechanism is plausible: the wristband uses two motors placed on opposite sides of the wrist, so left and right are encoded *spatially* (which side vibrates), while Slow Down and Obstacle are encoded *temporally* (different pulse patterns from the same motors). Spatial encoding requires no learning because it maps directly onto body schema; temporal encoding requires the user to count or time pulses, which is cognitively expensive and easily confused under attention load. RQ1 is therefore answered partially: directional cues are usable as designed, but the non-directional vocabulary needs to be re-encoded spatially (e.g., a third motor in a distinct location) before V4.

Vibration is perceivable but not yet operationally reliable (RQ2 Part A). Overall pretest accuracy of 72.5% was significantly above chance ($p < .0001$), confirming that vibration cues are perceivable through the wristband form factor. However, accuracy did not exceed the 75% HCI usability threshold ($p = 0.729$). The current motor intensity is sufficient as a baseline, but the system as a whole is not yet at a level we would call operationally reliable. Importantly, the gap between the two benchmarks is concentrated in the two non-directional signals: if Slow Down and Obstacle were

redesigned to perform like the directional cues, overall accuracy would clear the 75% threshold without any change to motor intensity. This reframes RQ2A: the bottleneck is signal vocabulary design, not raw perceptibility.

Movement does *not* degrade perceptibility (RQ2 Part B). This is our most consequential and counter-intuitive finding. Response times during the live moving session (M = 2.33 s) were significantly *faster* than during the stationary pretest (M = 4.93 s; mean difference 2.60 s, $t(39) = 3.574$, $p = 0.001$). Both participants confirmed this qualitatively: P2 reported no detectability difference, and P1 reported heightened tactile sensitivity ("without the sight, it made my touch more sensitive"). This contradicted the prediction from our formative study with the Dark Running Group that "vibration becomes imperceptible when running." We interpret the speed-up as an attention effect: live navigation demands continuous monitoring of haptic cues, while the pretest allowed slow deliberation. We cannot fully separate this from a possible confound — live-session response times measured reaction to a command in a flowing task, while pretest times measured deliberate identification of an isolated cue — but even with that caveat, the qualitative reports independently support the conclusion that walking-speed motion does not block perception. The practical implication is significant: the V3 vibration approach is worth refining rather than abandoning. We do *not* need to commit to force-based actuators in V4 purely on perceptibility grounds, freeing us to invest that effort in signal vocabulary redesign instead.

Trust is moderate and contextual (RQ3). Confidence ratings clustered at 6–7/10, with both participants explicitly conditioning their score on movement speed and route complexity (P1: "if I wasn't moving too fast"; P2: "if I had a more complex path"). This is internally consistent with the accuracy data: a system that misidentifies non-directional signals roughly half the time cannot yet justify higher trust, and users intuitively recognized this without being shown the numbers. The contextual framing also tells us *where* trust would grow: users implied that fixing the confused signals and validating at running speed are the levers, not adding new functionality. Notably, P2's unsolicited design suggestion (relocate the obstacle motor to the palm side so it doesn't conflict with the other sensors) is a direct user-generated solution to the Theme 2 confusion and gives us a concrete starting point for V4.

6 Limitation and Future Work

Limitations. This study has several important limitations that should be considered when interpreting the results. The extremely small sample size (n=2) substantially limits statistical power and reduces the ability to generalize the findings to a broader population. Because of the limited number of participants, even small differences in individual performance could strongly influence the statistical outcomes. The study relied on sighted blindfolded participants rather than individuals who are blind or have low vision (BLV). Although blindfolding can approximate the removal of visual input, it does not replicate the lived experiences, mobility strategies, or sensory feel developed by BLV users over time. As a result, the current findings may not accurately reflect how the intended user population would perceive or respond to the haptic commands. In addition, the live evaluation was conducted in an indoor walking environment instead of a realistic outdoor running context. Real-world deployment would likely involve additional distractions and challenges such as uneven terrain, environmental noise, weather conditions, and higher movement intensity.These could all affect reaction time and command recognition. Finally, the study focused primarily on short-term recognition accuracy and response time, but did not evaluate other important usability factors such as comfort, cognitive workload, learnability, or long-term adaptation to the haptic language.

Future Work. Future iterations of this project should focus on improving both the experimental design and the haptic feedback system itself. Most importantly, future studies should recruit pBLV runners rather than blindfolded sighted participants in order to better evaluate how the intended user population interprets and adapts to the haptic commands. Since experienced BLV users may possess stronger tactile sense and greater familiarity with nonvisual navigation cues, their performance could differ highly from the results observed in this pilot study. In addition, the Slow Down and Obstacle Warning commands should be redesigned to include more distinct and dynamic haptic patterns. Participants said there were confusion between these commands, suggesting that the current vibration patterns were too similar under movement conditions. Increasing the contrast between signals through modifications in vibration intensity, pulse duration, rhythm, or sequencing could improve recognition accuracy and reduce response time. Future experiments should also incorporate larger sample sizes and repeated testing sessions to better measure consistency, learning effects, and user adaptation over time. Expanding testing into outdoor running environments would provide a more realistic evaluation of system performance under real deployment conditions, including environmental distractions and varying terrain.

7 Conclusion

People who are blind or have low vision (pBLV) continue to face barriers to independent outdoor running due to safety concerns and reliance on sighted guides. To address this gap, we developed the Virtual Tether system, a wearable assistive navigation concept that combines real-time computer vision with non-visual haptic feedback. Through a human-centered iterative design process, we conducted formative interviews, developed three progressively refined prototypes, and integrated them into an end-to-end wearable navigation system evaluated through a pilot study with two blindfolded sighted participants.

The pilot evaluation produced several key findings. Overall accuracy reached 72.5%, demonstrating that vibrotactile cues are perceivable and usable above chance levels. Directional signals, particularly Left Turn and Right Turn, performed strongly both quantitatively and qualitatively, while Slow Down and Obstacle Warning were frequently confused, revealing limitations in the temporal encoding of non-directional cues. One of the most significant findings was that response times during movement were significantly faster than during the stationary pretest, contradicting the initial expectation that vibration feedback would become harder to perceive while moving. Participants also reported that the haptic feedback remained noticeable during walking, supporting the viability

of vibration-based navigation for active movement contexts. Overall, this work demonstrates the feasibility of combining wearable sensing and haptic feedback to support more independent outdoor navigation for pBLV runners while also identifying important design improvements needed before real-world deployment.

This work contributes an initial HCI baseline for wearable, non-visual navigation systems designed specifically for independent outdoor running among pBLV users, an area that remains largely unexplored in prior research. Through an iterative human-centered design process and pilot evaluation, we demonstrate that real-time vibrotactile guidance during movement is both feasible and perceptible, establishing a foundation for future assistive running technologies and haptic interaction research in high-mobility contexts.

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